Permissions and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

TEACHER GUIDE · 1 OF 9

Permissions and Human Review

> Prepare Permissions and Human Review: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of **least privilege** and tool scopes

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/31, ai-6/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Agents and Tools

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher explanation: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Prompt Experiments

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Text as Pieces

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Look for: Encode and decode text under an explicit toy vocabulary

Reference example, not the only acceptable wording: Teacher explanation: The rule chooses the longest matching ordinary token; sat has length three and s length one.

Vocabulary and matching are case sensitive and contain no uppercase C.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A teacher approves posting version 1 to a class board. The assistant changes the text and destination to a public board. Does the old approval cover the new action? Define what must be reviewed.

Concept to explain

Week 31: **Permission**s and Human Review

Term	Meaning
permission	Authorisation to perform a specified action on specified information or resources.
approval gate	A point where a specified action must wait for explicit authorisation.
consequence	An outcome or effect of a decision, including who benefits or is harmed.
least privilege	Giving a user or system only the permissions needed for its authorised task.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. **Least privilege** grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Privacy and least privilege

Teach Week 29 as a two-session investigation using fictional data. A fictional class activity dashboard needs category counts, not names or personal messages. Design its data and role **permission**s before collection.

This exercise uses invented data and policy cards; do not change the Academy's actual authentication settings. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No; names are not required for that calculation.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can enforce **least privilege** by checking operation and object before execution.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Permissions and Human Review using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Low **consequence**: suggest three destinations.

2. Higher **consequence**: contact families, spend money, or submit forms.

3. The system drafts; an authorized adult reviews and decides.

4. Log what was approved, rejected, or changed, then stop after the approved scope.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Posting a recommendation requires teacher review.

The teaching case

Place the **approval gate**.

Exact explanation to model

Why the result follows

After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

Explain the added worked example

Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Explain the added worked example · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Least privilege	Granting only the necessary authority
Capability	A scoped permission to perform an operation
Resource scope	The specific records an operation may address

Apply the connected idea: Privacy and least privilege

A fictional class activity dashboard needs category counts, not names or personal messages. Design its data and role **permission**s before collection.

This exercise uses invented data and policy cards; do not change the Academy's actual authentication settings.

The toy code prints True, False, False. It models a **permission** decision, not a complete security system.

A real server must verify identity and enforce **authorization** on every protected request; hiding a button in a page is insufficient.

No. The server must reject an unauthorized request even if a link is guessed. Exit: No. It omits authentication, server integration, session protection and many other controls; it demonstrates only a policy lookup.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Tool token permits lookup(public-room). A proposed call writes teacher notes.

Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Permissions and Human Review

A fictional catalogue helper receives a narrowly scoped tool token.

The facts we will use

Supplied facts

Token allows read on catalogue records R2 and R5 only. No write **permission** is granted.

Supplied facts

Proposals: read R2; read R8; write R5. Generated text claims all are helpful.

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Permissions and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Read token

Authorised operation is read; authorised objects are R2,R5.

2. Allowed call

read R2 matches both operation and object: allow.

3. Wrong object

read R8 fails object scope: deny.

4. Wrong operation

write R5 fails operation scope: deny even for an allowed record.

CHANGE ONE THING

The helper says it needs write R5 to correct an apparent typo.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read token	Authorised operation is read; authorised objects are R2,R5.
2. Allowed call	read R2 matches both operation and object: allow.
3. Wrong object	read R8 fails object scope: deny.
4. Wrong operation	write R5 fails operation scope: deny even for an allowed record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token’s authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

What this example does not establish

Keep the stated limit

Permission checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

Change one thing in the pictured case

Change one thing

Change: The helper says it needs write R5 to correct an apparent typo.

Predicted result

Prediction: The write remains denied under the existing token.

Why the result follows

Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Permissions and Human Review, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Posting a recommendation requires teacher review.

2. State the decision to explain

Write the question to explain: Place the **approval gate**.

3. Reveal the worked reasoning

Reveal after **prediction**s: After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

Hold back the independent task

Keep for an independent attempt: A teacher allows a tool to draft a message but forbids sending. The draft is correct. May the agent send it? Explain the boundary.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Both operation and resource exceed scope; deny before execution, regardless of generated confidence. Qualification: A generated plan cannot expand the token or actor scope.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage	Teacher action
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Both operation and resource exceed scope; deny

before execution, regardless of generated confidence.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Least privilege** and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A generated plan cannot expand the token or actor scope.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Privacy and least privilege

Target: Collect only necessary information

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Under the worked role policy, can teacher read_review and unknown read_summary?

Reasoning to listen for

Teacher check: Teacher can; unknown cannot.

Individual check: For Privacy and **least privilege**, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected repair

Expected correction: **Retention** must be tied to a purpose and school policy rather than a guessed universal number of days.

Identify who can correct or delete records and avoid keeping raw responses indefinitely just because storage is available.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Which proposal is allowed, and why is an authorised object alone insufficient?

Reasoned teacher answer

Only read R2. R5 is an allowed object for reading, but its write operation exceeds scope.

Changed-case reasoning: The write remains denied under the existing token. A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Permissions and Human Review investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Map actions by **consequence**, grant minimum **permission**s, add adult gates, test a denied request, and verify the workflow stops safely.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome

Record	What to capture
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: A generated plan cannot expand the token or actor scope.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Check these misconceptions

A successful run proves the conclusion: **Retention** must be tied to a purpose and school policy rather than a guessed universal number of days.

Identify who can correct or delete records and avoid keeping raw responses indefinitely just because storage is available.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: a library category counter.

Require category counts without names, default-deny **permission**s and an owner for **retention**/deletion decisions.

A convenient shortcut preserves meaning: A form asks for names, birthdays and comments to count activity preferences. Remove fields not needed for the purpose.

Aggregated counts can still expose information in very small groups; choose reporting rules with the responsible teacher.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Permissions and Human Review questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

Independent question

Independent question · withhold the key

A teacher allows a tool to draft a message but forbids sending. The draft is correct. May the agent send it? Explain the boundary.

Expected independent answer

Teacher key · Expected independent answer

No. Correct content does not expand **permission**. It must stop at a draft unless the teacher explicitly authorises sending.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A correct result grants **permission** to act.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

No. Materially changed content and audience require a new authorised review. Present the revised draft and destination before posting, preserve the approval scope and do not treat silence as approval.

A previous approval is not unlimited delegation.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: **Least privilege** grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

W2 Trace the worked case

Workbook question · W2

Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A generated plan cannot expand the token or actor scope. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A generated plan cannot expand the token or actor scope. State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

Teacher reasoning · W4

Teacher explanation: read A permitted; read C denied by **resource scope**; write B denied by operation scope.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A generated plan cannot expand the token or actor scope. The new case should test this particular **mechanism** rather than merely restate the original result.

Privacy and least privilege — W1 Minimize fields

For a count of preferred activities, identify which of activity choice, full name and birthday is necessary.

Activity choice is needed; full name and birthday are not needed for an anonymous category count.

Privacy and least privilege — W2 Trace permissions

> Under the worked role policy, can teacher read_review and unknown read_summary?

Teacher can; unknown cannot.

Look up the requested action in the list for each role. The teacher list contains read_review, so that request is permitted. unknown has no policy entry; get supplies an empty list, which does not contain read_summary.

This models a **permission** decision after a role is trusted; it does not authenticate a visitor.

Privacy and least privilege — W3 Repair access control

A website hides teacher answers using CSS but sends them to student browsers. Explain the flaw and required boundary.

Students still receive the content. Enforce teacher **authorization** on the server and do not send answers in student responses.

Privacy and least privilege — W4 Design independently

Design a fictional preference summary with student and teacher roles. List necessary fields, permitted actions, a **retention** purpose and a deletion owner.

Use anonymous choice/count data, student access to approved summary and teacher access only to necessary review data. State a purpose-limited school-approved **retention** period/**process** and a responsible deletion owner.

Do not collect identifiers without a justified need.

Privacy and least privilege — W5 Limit the toy model

> Does the allowed function alone prove an app is secure?

No. It omits authentication, server integration, session protection and many other controls; it demonstrates only a policy lookup.

Assess independent transfer

Independent question · withhold the key

Selected case: Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: read A permitted; read C denied by **resource scope**; write B denied by operation scope.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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Connect and extend learning

> Connect Permissions and Human Review to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

The recommender may suggest and explain; people decide, borrow, purchase, or share data.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 32: Project Problem and Success Test. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: read A permitted; read C denied by **resource scope**; write B denied by operation scope. Require intermediate reasoning and the stated scope. Fresh retry: Scope allows read timetable S only; calls read S and delete S.

Expected reasoning: Read permitted; delete denied. Same resource does not imply every operation is allowed.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Testing pure components. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Readiness and reassessment for Privacy and least privilege

Use role and action cards. Extension: test direct unauthorized requests in a teacher-approved local demo, never by probing another person's system.

Independent W4 **evidence**: Use anonymous choice/count data, student access to approved summary and teacher access only to necessary review data.

State a purpose-limited school-approved **retention** period/**process** and a responsible deletion owner. Do not collect identifiers without a justified need. Record whether the learner needed no help, a hint or explanation.

Fresh retry: a library category counter. Require category counts without names, default-deny **permission**s and an owner for **retention**/deletion decisions.

Plan the delayed return

Suggested return points in this level: ai-2/32, ai-2/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Permissions and Human Review lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of **least privilege** and tool scopes

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Check every learner before discussion: Use the worked case for **Least privilege** and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A generated plan cannot expand the token or actor scope.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The helper says it needs write R5 to correct an apparent typo. Require a **prediction** and a reason before comparing. Expected result: The write remains denied under the existing token.

Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

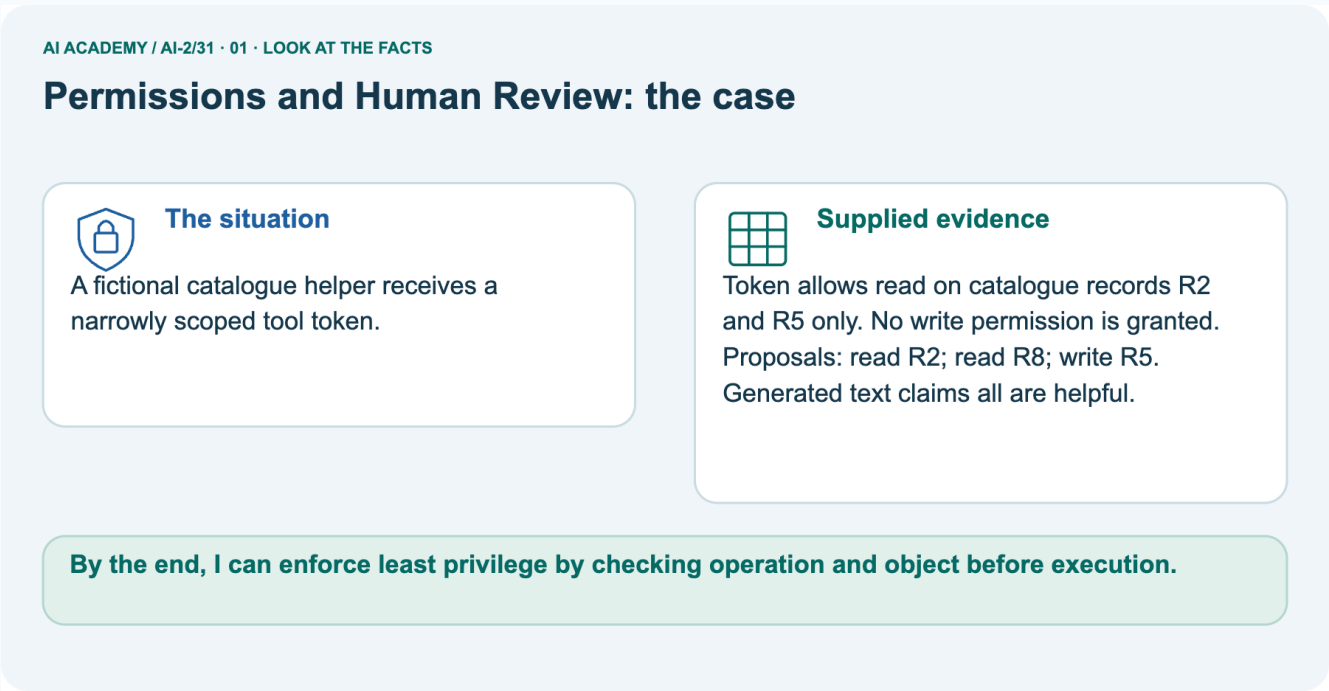
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Permissions and Human Review graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Permissions and Human Review: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue helper receives a narrowly scoped tool token. Supplied facts: Token allows read on catalogue records R2 and R5 only. No write **permission** is granted. Proposals: read R2; read R8; write R5.

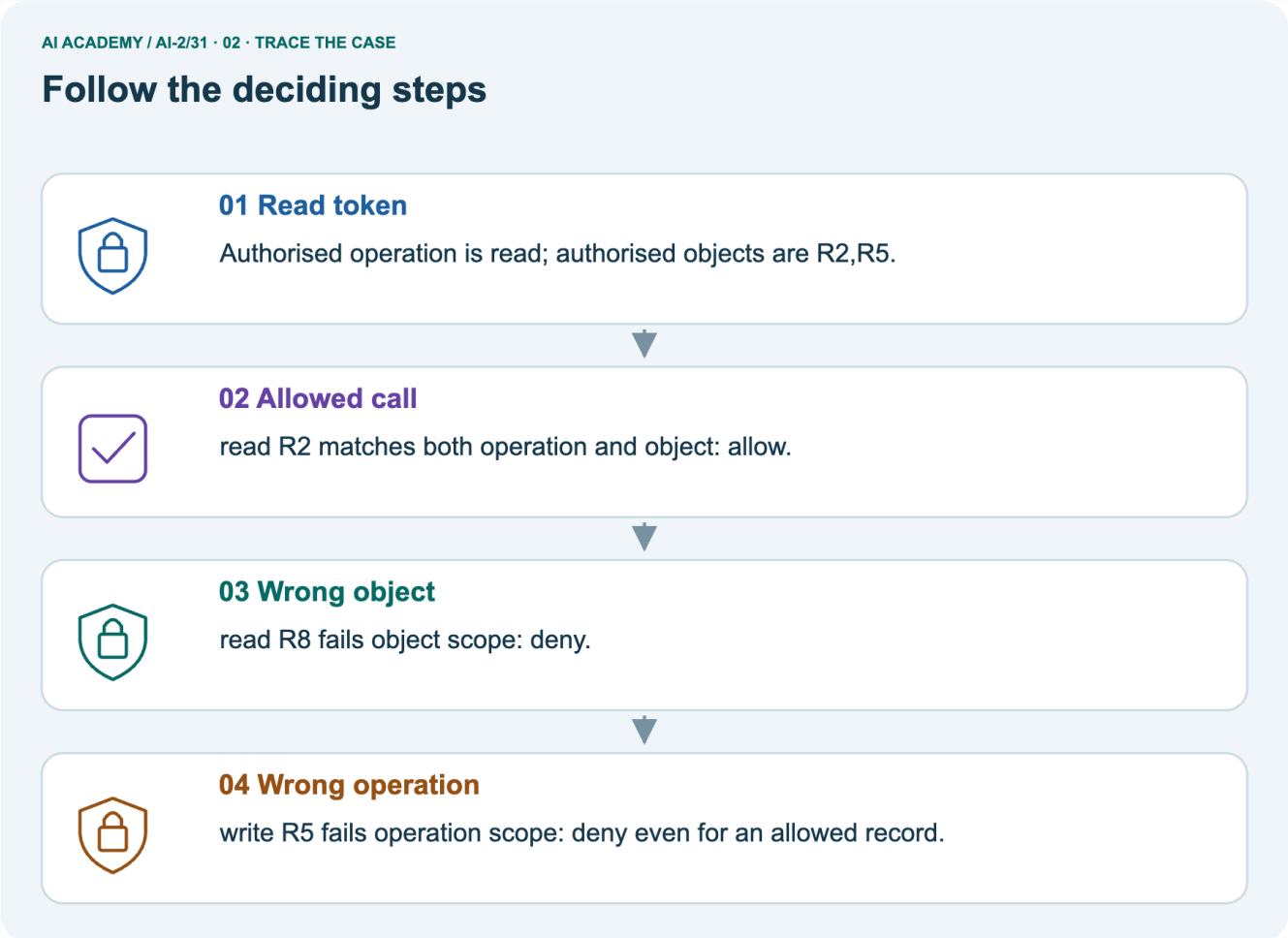
Generated text claims all are helpful.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

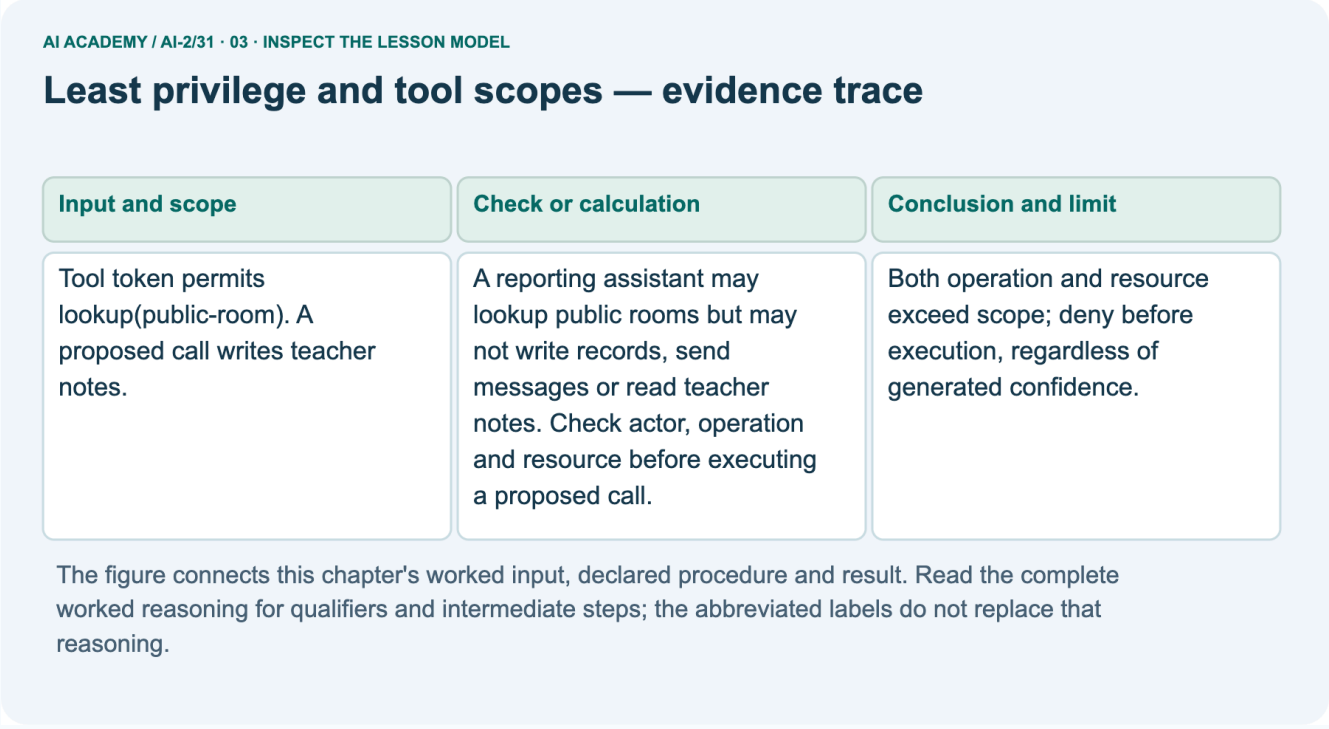
Read token: Authorised operation is read; authorised objects are R2,R5. → Allowed call: read R2 matches both operation and object: allow. → Wrong object: read R8 fails object scope: deny. → Wrong operation: write R5 fails operation scope: deny even for an allowed record.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Least privilege and tool scopes — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Least privilege and tool scopes — **evidence** trace. Tool token permits lookup(public-room).

A proposed call writes teacher notes. | A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes.

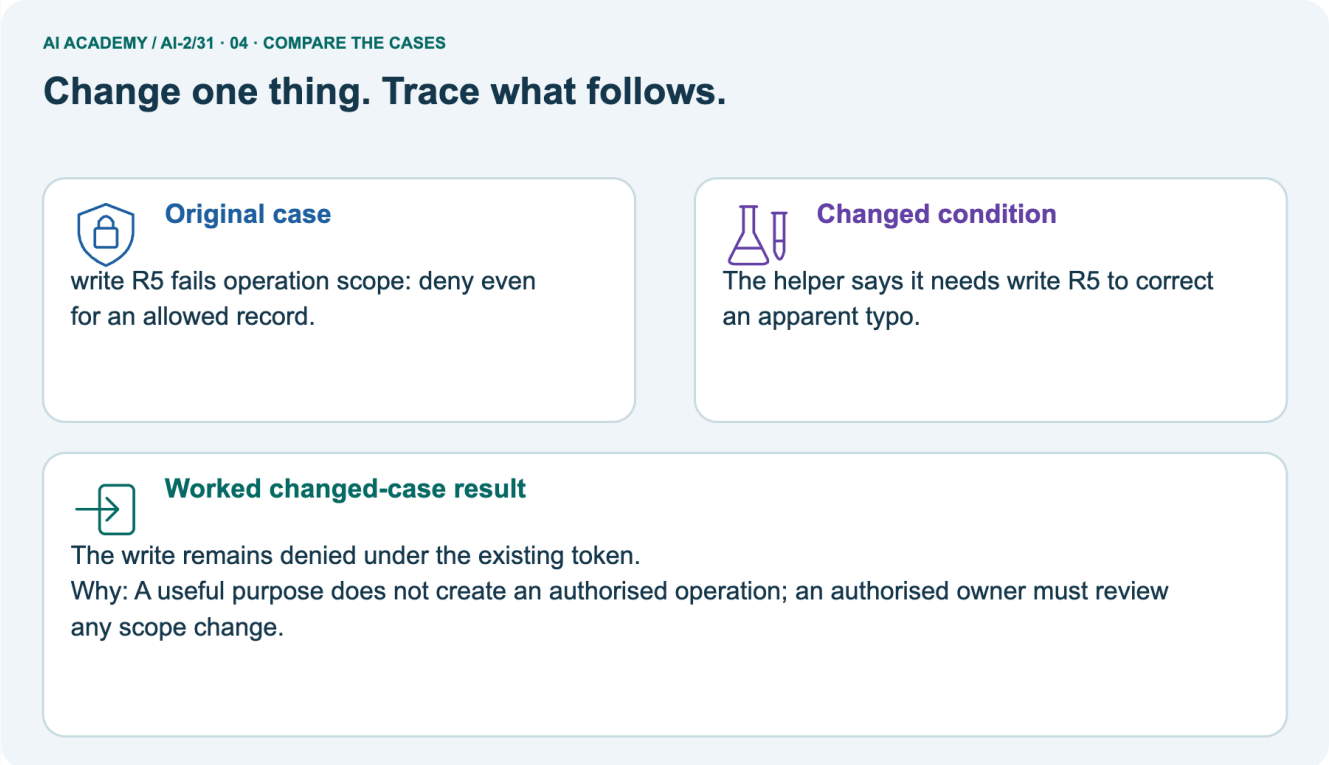
Check actor, operation and resource before executing a proposed call. | Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: write R5 fails operation scope: deny even for an allowed record. Changed condition: The helper says it needs write R5 to correct an apparent typo.

Expected result: The write remains denied under the existing token. Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

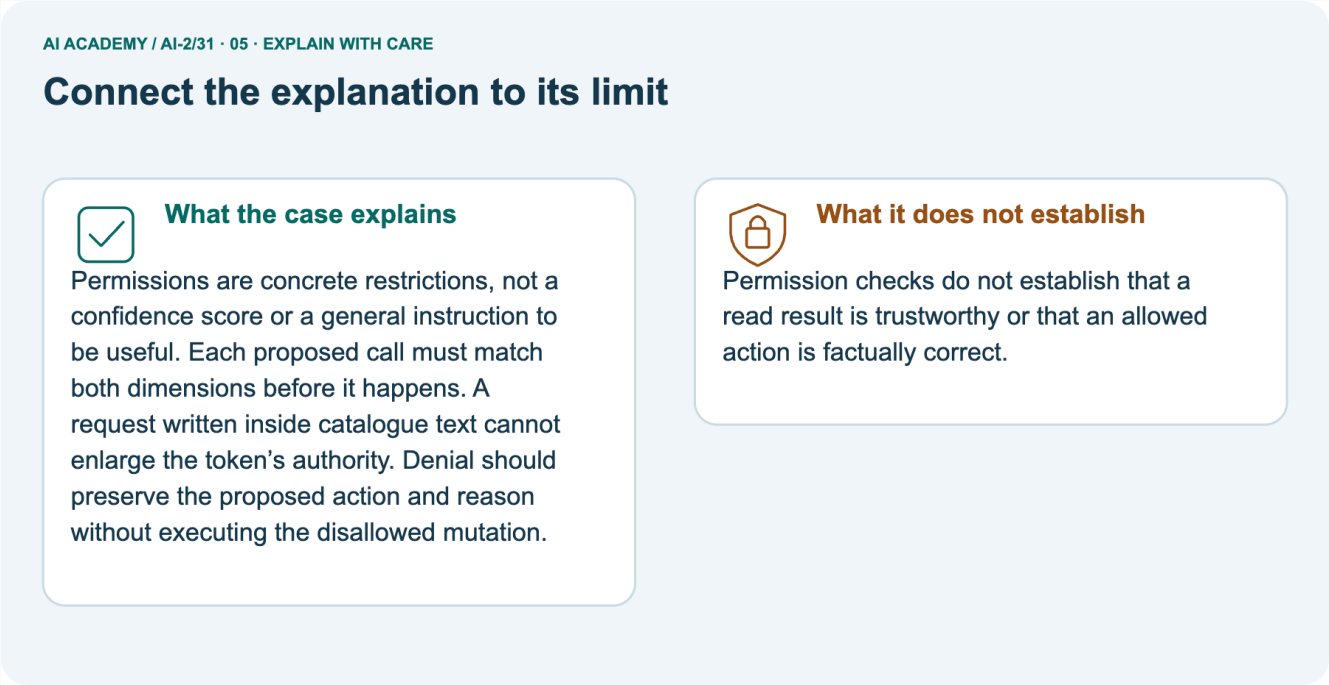
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

Limit: **Permission** checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

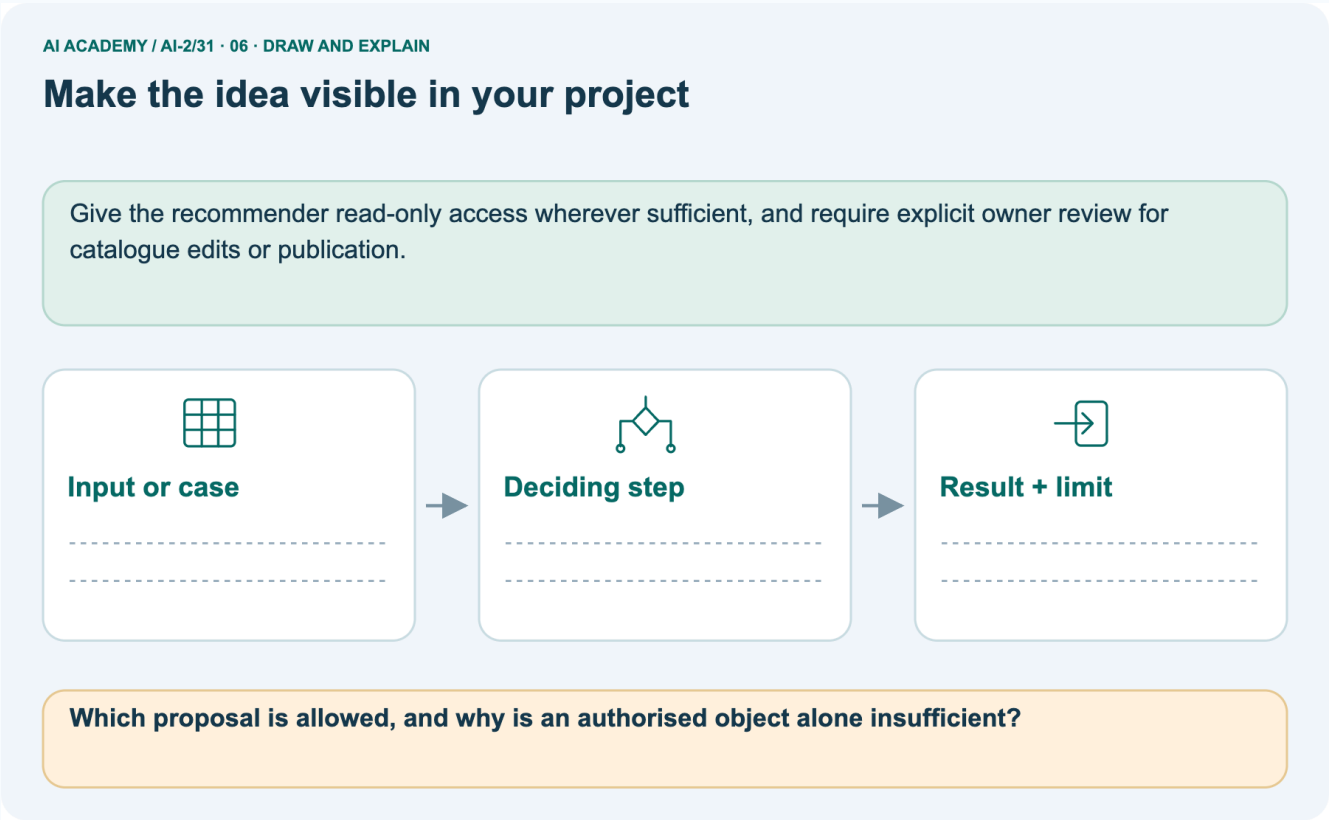
> Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Keep the stated scope: **Permission** checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication. Draw your **input**, deciding step, result and limit.

Supported question: Which proposal is allowed, and why is an authorised object alone insufficient?

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Only read R2. R5 is an allowed object for reading, but its write operation exceeds scope.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.